

CryptoAgent URL:

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CryptoAgent Project Part B: Model Design and Implementation*

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Abstract

This report introduces CryptoAgent, a framework for cryptocurrency investment designed to handle the market's high volatility. CryptoAgent is a FinTech framework that blends hard data and market sentiment, my goal is to help investors cut through the noise by combining quantitative factor investing with advanced sentiment analysis, make decision that are not just based on numbers, but also on how people actually feel and react in real time. Further, The system is conceptualized as a multi-agent framework, where specialized "agents" for quantitative analysis, sentiment monitoring, and risk management collaborate to inform a final investment strategy.

Keywords: Crypto Sentiment Analysis, VADER Lexicon Expansion, Multifactor Portfolio, Behavioral Biases, Crypto Portfolio Optimization

1 Introduction

Cryptocurrency has become a significant asset class, but it is also known for its high risk and extreme price fluctuations. For example, by December 2024, the price of Bitcoin had risen back to \$100,000 after falling from over \$66,000 at the end of 2021 to just over \$16,000 in early 2023.(Li et al. 2025) Unlike the traditional stock market, the price of crypto is primarily driven by investors’ perceptions of its growth potential. This environment often leads to emotional trading decisions, such as ”Fear of Missing Out” (FOMO) during price rallies and panic selling during downturns. However, the previous studies typically focus only on the impact of sentiment on price, often overlooking the reverse effect where price movements will shape sentiment. Negative news can instill fear and cause selloffs, but a sudden price drop can also trigger anxiety and generate negative news coverage.

2 Station 3 Model Design and Sentiment Analysis

This section provides a detailed breakdown of the models and methodologies that form the analytical core of CryptoAgent. Basically, station 3 is where we turn all our input into an actual trading model. It takes the two signals we created earlier: (1) signals from historical prices, like whether a crypto is in an uptrend, and (2) our score that tells us the real mood of the market based on validated news. Therefore, we employ two types of analysis: one is a sentiment analysis model that helps us understand the stories shaping the cryptocurrency market. The other is a set of quantitative portfolio models that track key market trends.

2.1 VADER

Our primary tool for analyzing news text is the VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment model, chosen for its suitability to our product. According to Hutto and Gilbert (2014), VADER is a transparent, rule-based system that is both computationally efficient and highly interpretable. This is a critical advantage in our product, because understanding why a model makes a certain decision is as important as the decision itself. Furthermore, VADER is based on a human-validated lexicon; the author constructs and empirically validates a gold-standard list of lexical features, specifically tailored to sentiment in microblog-like contexts. For instance, correctly investing sentiment in phrases like “not a positive chance” is crucial, as is recognizing that “extremely bullish” carries more weight than “bullish”. Third, correctly interpreting that in a sentence like “The food here is great, but the service is horrible”, the clause after but dominates the overall sentiment.

2.2 VADER’s Limitation in the Crypto Market

Despite VADER’s potential, it is not without limitations, particularly when applied to a specialized domain like the cryptos market. Because VADER’s effectiveness is directly tied to its dictionary’s coverage, excluding the crypto-specific jargon, VADER might misinterpret or entirely miss the sentiment

in a news article. For instance, important terms like “rugpull”, “bullish”, “bearish”, and “airdrop” are missing, which may cause CryptoAgent to miss significant sentiment signals. Second, VADER may misinterpret words like “liquidations” or “resistance,” which have neutral general meanings but carry strong negative sentiment in a trading context.

2.3 Lexicon Expansion

Considering the VADER’s limitations, we expand the VADER customer lexicon module, which contains a dictionary of cryptos and financial terms. We are adding over 50 terms to the current lexicon, structured into logical categories to ensure comprehensive coverage. Tab.1 provides a sample of our custom terms.

Table 1: Key Table – Sample Add-on Crypto Sentiment Lexicon

Term	Score	Rationale (why the polarity)	Suggested Source
spot_etf	2.6	SEC approval / filing of a spot-based ETF is viewed as a major step toward main-stream adoption.	Crypto/Finance
proof_of_reserves	1.7	Exchange transparency boost → increases depositor confidence.	Crypto Ops
node_outage	−1.9	Chain liveness halted; raises reliability concerns.	Tech Incident DB
chain_reorg	−2.7	>1-block rollback threatens finality & can enable double-spend.	Consensus Research
token_unlock	−1.3	New vested supply hitting the market → potential sell-pressure.	Tokenomics Docs

After the model analyzes a piece of text and applies all its grammatical rules, each word, emoticon, and slang term is assigned an adjusted sentiment score, which we will denote as s for a given token. The positive, negative, and neutral scores are calculated based on the proportion of the text’s sentiment that falls into each category. Eventually, output the compound score, which consolidates all token scores into a single score for the entire news. First, s is calculated by simply summing the adjusted sentiment scores of all tokens in the text: $S = \sum s$. This sum is then normalized using the following equation to produce the final compound score:

$$\text{compound} = \frac{S}{\sqrt{S^2 + 15}}$$

In this formula, the 15 represents the normalization factor that keeps the output between -1 and +1. The function forms an S-shaped curve, so when the raw sum s is very high or low, the compound score gets closer to ± 1 . This score serves as our primary input for the next steps in the analysis. Each pre-processed news article is scored using our customer VADER model. The pipeline uses two filters: a sentiment threshold $|\text{compound}| \geq 0.05$ removes neutral articles, and second, it removes articles that are shorter than the daily 25th percentile in word count, which helps exclude content of low substance. Furthermore, the pipeline implements a substance weighting scheme to prevent giving undue influence to short articles, allowing longer and more detailed articles to have a greater impact on the final

sentiment index. Thus, creates both daily and weekly time series for the overall market and for each major cryptocurrency tag. These indices are saved as CSV files and are used in our trading strategies and analysis.

2.4 Result and analysis

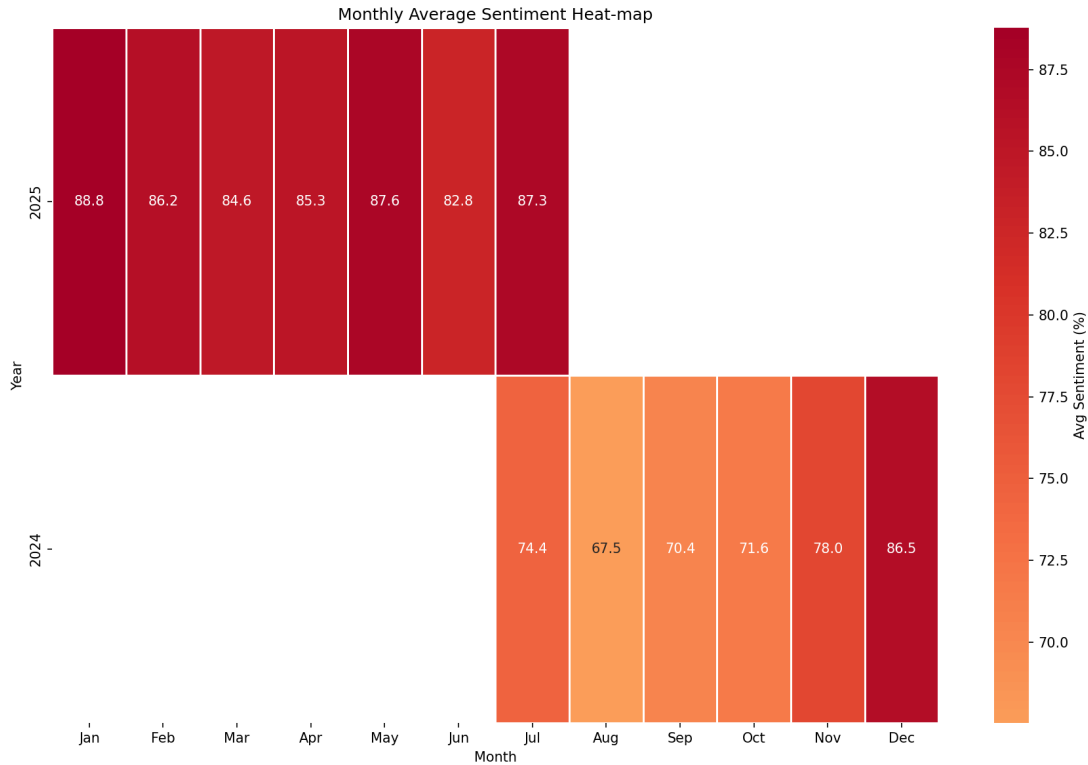
First, we needed to check if our model’s positive or negative classifications were accurate. Tab.2 shows how well the model distinguishes between positive (class 1) and negative (class 0) news articles.

Table 2: Classification report

Class / Avg	Precision	Recall	F1-score	Support
0	0.86	0.34	0.49	9523
1	0.70	0.96	0.81	14 903
Accuracy			0.72	24 426
Macro avg	0.78	0.65	0.65	24 426
Weighted avg	0.76	0.72	0.68	24 426

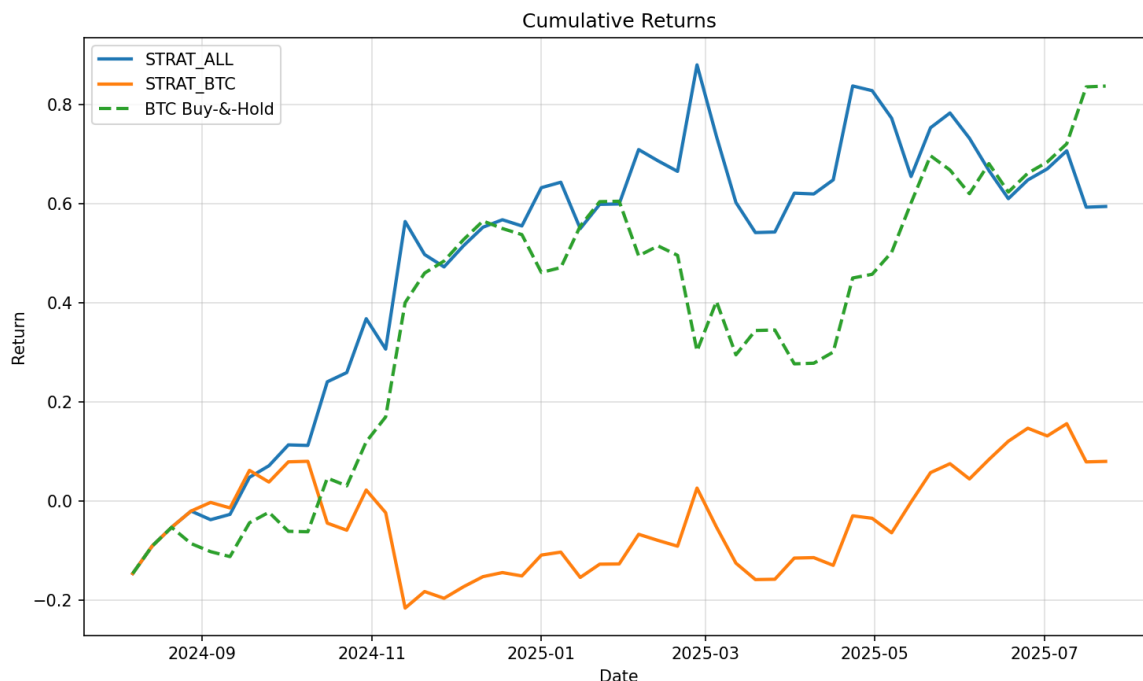
The model demonstrates a very high recall of 0.96 for positive news from CoinDesk, indicating that it successfully identifies 96% of all positive news. However, the recall for negative news is significantly low at 0.34, suggesting that our model may fail to identify nearly two-thirds of negative news. Second, to provide context for the backtest, we aggregate sentiment by month. (Shown in fig.1)

Figure 1: monthly_avg_heatmap



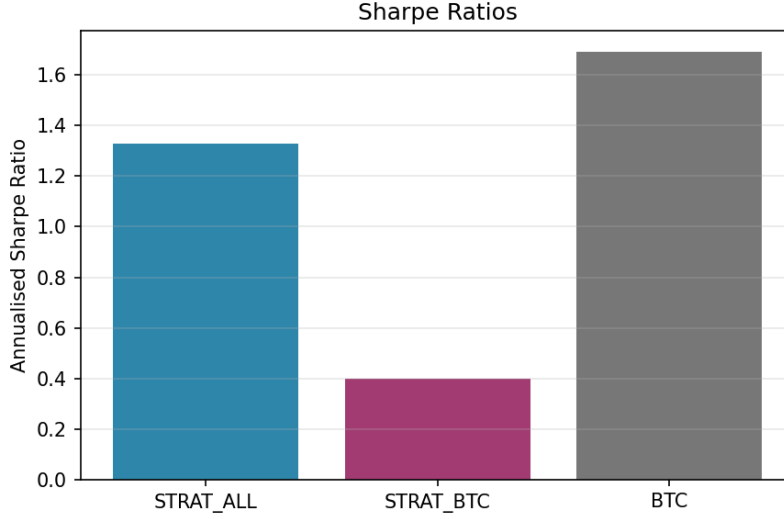
The heat map indicates that the overall sentiment improves from the second half of 2024 to most of 2025, revealing that sentiment may exhibit persistence at monthly horizons and is suitable for weekly trading rules. Third, the all-tags sentiment strategy (blue line) performed quite well, tracking the performance of holding Bitcoin closely, but with a smoother ride, during the market dip in early 2025. (Shown in pic. 2)

Figure 2: Cumulative Returns of Sentiment Strategies vs. BTC Buy-and-Hold



On the other hand, the BTC only (**STRAT_BTC**) lags for most of the window. This reveals a fact that using sentiment from a broad range of news articles is more effective than focusing only on news about a single cryptocurrency. Finally, risk-adjusted results confirm that buy-and-hold BTC achieves the highest Sharpe over this sample, but **STRAT_ALL** attains a materially higher Sharpe than **STRAT_BTC**. (Shown in fig.3)

Figure 3: Sharpe Ratios of Sentiment Strategies



3 Quantitative Factor Portfolio Models

In parallel with our sentiment analysis, the CryptoAgent framework includes a suite of quantitative strategies. Our quantitative models comprise a set of investment factors derived from daily price and volume data from Station 2, including momentum, volatility, and short-term reversal. First, momentum is based on the research conducted by (Jegadeesh & Titman 1993), which focuses on buying stocks that are outperforming and selling those that are underperforming. Moreover, we calculated momentum over multiple lookback periods (e.g., 14, 21, 28, 42 days) to test its effectiveness across different timeframes. Second, we utilize a volatility factor that aims to provide investors with a smoother ride within equity allocations by creating a portfolio that exhibits fewer swings than the cryptocurrency market. Third, we included a short-term reversal factor, which tests the idea that if a crypto experiences an extreme price jump following a one-week drop, it's likely to move in the opposite direction the next week.(Lehmann 1990)

The next step in our process was to transform these abstract signals into concrete investment strategies. This entire portfolio construction and backtesting procedure is handled in Station 3 by using PyBondLab. Our method involves a weekly rebalancing process based on a cross-sectional sorting approach. Specifically, we rank all cryptocurrencies by their latest factor scores and partition the ranked universe into five equal-sized portfolios. Moreover, Q5 has the. The ranked asset, while Q1 contains the lowest-ranked assets, allows us to compare those that strongly display a trait to those that do not.

3.1 Dynamic Factor-Weighting Model

Instead of trying to guess which single factor will perform best, we built a dynamic model that allocates our investment across all the base strategies every single week. My ideas aim to automatically adapt to changing market conditions. In more detail, every week, we review the performance of each of our

base strategies over the last 12 weeks and calculate a score for each, primarily looking for strategies that have delivered the best returns. The formula for this score, S , is:

$$S_{i,t} = \frac{\mu_{i,t}^{(12)}}{\sigma_{i,t}^{(12)}}$$

Second, we determine how much money to allocate to each strategy for the upcoming week based on its score. Then we use a function call softmax, which allows us to smartly assign more weight to the strategies with a higher score recently, while less weight to the other strategies:

$$\tilde{w}_{i,t} = \frac{\exp(S_{i,t}/\tau)}{\sum_j \exp(S_{j,t}/\tau)}$$

Furthermore, we added two common-sense rules: first, we cap the amount of money that any single strategy can receive at 60% of the entire portfolio. Second, we limit the amount of change in a strategy's allocation from one week to the next to 20%. Finally, we calculate the return of our combined portfolio, making sure to subtract an estimate for transaction costs. We assume a cost of 20 points for the portion of the portfolio that we trade each week. The final return calculation is:

$$R_{t+1}^{\text{combo}} = \sum_i w_{i,t} R_{t+1}^{(i)} - \text{bps} \times \sum_i |w_{i,t} - w_{i,t-1}|$$

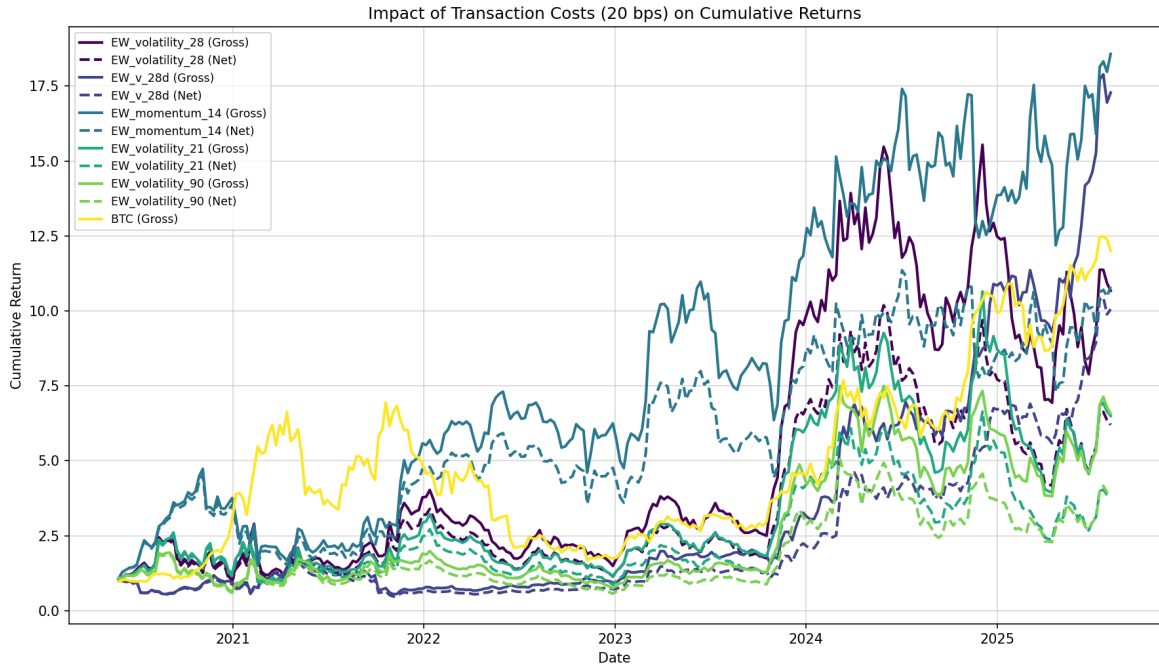
3.2 Transaction cost

We assumed a round-trip cost of 20 basis points (0.20percent) for every unit of the portfolio that is rebalanced each week, which reflects that strategies with higher turnover are penalized more heavily, and considers real-world implementation challenges.

3.3 Results and analysis

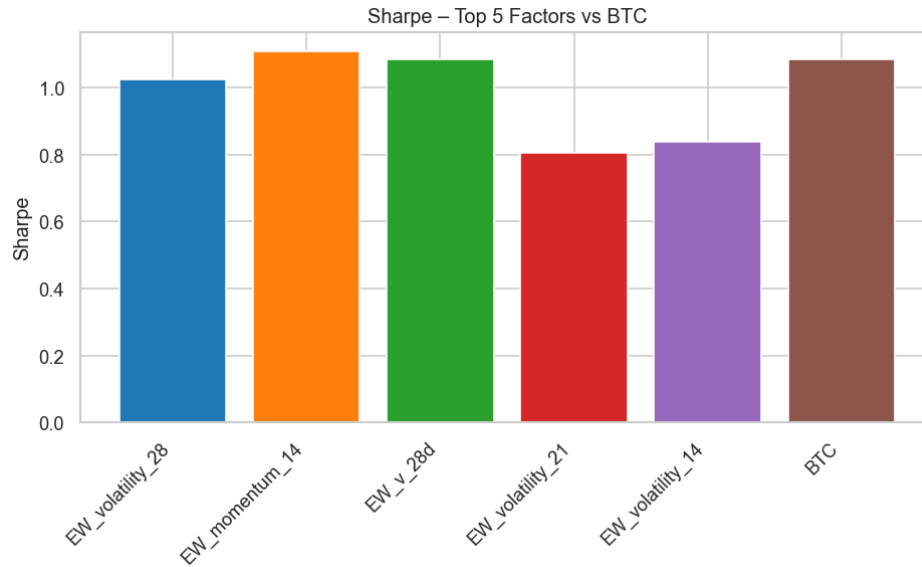
This section presents the results for our quantitative investment strategies, moving from the performance of the baseline single strategy to our dynamic allocation strategy. The analysis is based on a weekly backtest spanning from 2020 to 2025. Fig.4 displays the cumulative. Return of the top-five performing equal-weighted factor strategies, benchmarked against a BTC buy-hold investment.

Figure 4: Impact of Transaction Costs on Cumulative Returns



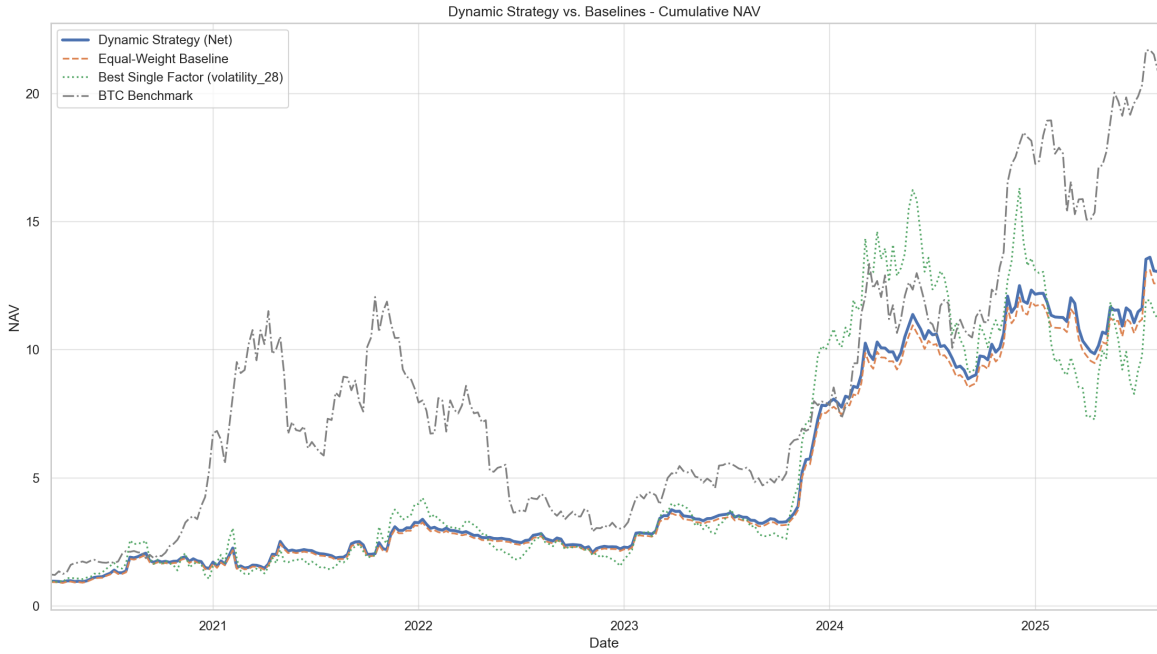
The results show the 14-day momentum factor (`EW_momentum_14`) emerges as the standout performer, achieving the highest cumulative return and outperforming the BTC benchmark over the full sample period. Second, the 28-day low volatility factor (`EW_volatility_28`) also demonstrates strong performance, suggesting that risk-averse strategies can also yield compelling returns. Second, fig.5 shows the annualised weekly Sharpe ratio ($\text{Mean} \times 52, \text{stdev} \times \sqrt{52}$). Results indicate that momentum (14 days and 28 days) both post Sharpe above 1.1, narrowly ahead of a BTC benchmark (1.08).

Figure 5: Annualized Sharpe Ratios



Moreover, Fig.6 presents the cumulative net-asset value (NAV) of the dynamic overlay compared with three anchors: an equal-weight blend of all factor legs, the EW volatility-28d, and a buy-and-hold position in BTC. Over the entire sample period, our approach almost aligns with or exceeds the equal-weight approach. From mid 2023, it pulls ahead and finishes the period with a NAV of $12\times$, versus $11\times$ for the baseline. Although the EW volatility-28d (green line) explodes higher in two sharp bursts, it also suffers mean reversion by the end of 2025. During periods where the single factor underperformed (such as in late 2021 and mid-2023), our approach achieves a steadier climb precisely because it reallocates away from legs whose momentum is fading and toward those that are taking the lead.

Figure 6: Cumulative Net Asset Value (NAV) Chart



Therefore, our model was able to mitigate losses by rotating into other, more stable and better-performing factors, resulting in a more stable and ultimately higher cumulative NAV. The tab. While the Bitcoin benchmark achieved the highest Sharpe ratio, it also had the Maximum drawdown (-75.90%), which is nearly double that of our dynamic strategy (-38.47%). In real-world investment, managed risk is much important than return. In addition, the single factor shows volatility was extreme (86.82%), and its maximum drawdown was a severe -62.96%. Conversely, our approach shows less volatility and achieves a 1.20 Sharpe ratio.

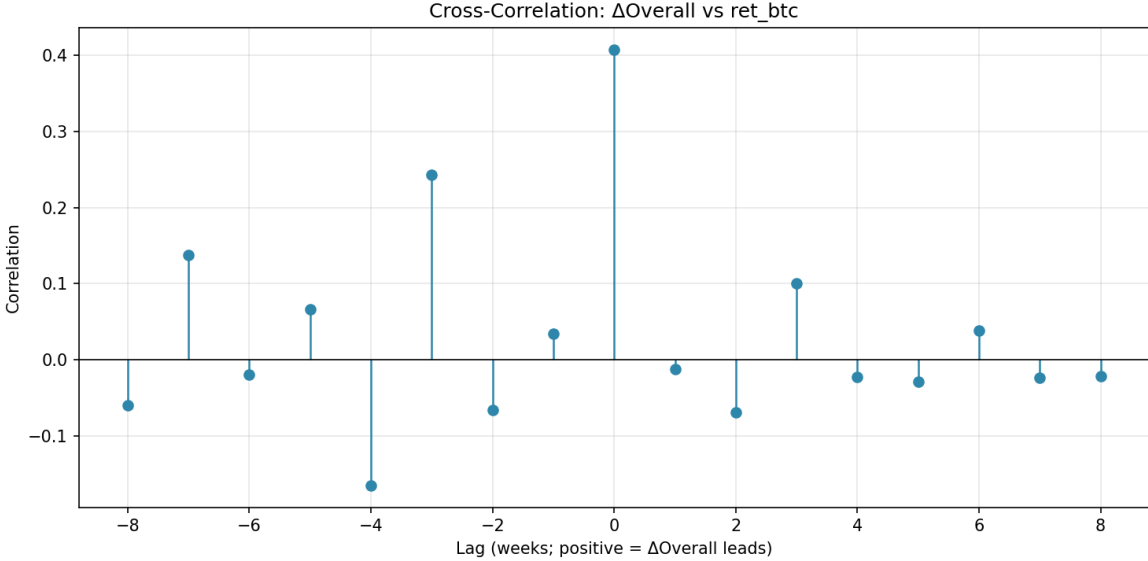
Table 3: Strategy Performance Summary

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Max Drawdown (%)
Dynamic Strategy (Net)	59.54	49.80	1.20	-38.47
Equal-Weight Baseline	58.89	49.91	1.18	-38.47
Best Single Factor (vol.28)	79.75	86.82	0.92	-62.96
BTC Benchmark	74.67	60.78	1.23	-75.90

3.4 Causality

Having analysed the performance of our sentiment and quantitative models independently, this section will briefly discuss the relationship between them and use these insights to inform the final design of our CryptoAgent. Fig.7 shows the lead-lag relationship between overall sentiment change and BTC returns, with a correlation of +0.41 at lag 0, indicating that positive sentiment and positive returns tend to occur within the same week.

Figure 7: Cross-Correlation Plot



However, in other lags, a weak trend is shown that sentiment might predict future prices. Moreover, the tab.4 Demonstrate that sentiment today does not immediately move the BTC price, but about three weeks later, there is a small uptick in returns (correlation = +0.16). Which meant sentiment cannot predict future returns.

Table 4: Lagged Correlation Between Sentiment Changes and Bitcoin Returns

Sentiment Metric	Lag 1	Lag 2	Lag 3	Lag 4
d_Overall	-0.01	-0.07	0.10	-0.02
d_BTC	-0.07	-0.03	0.16	-0.04

4 Model Implementation

Station 4 represents the culmination of previous development stages by integrating all models into a cohesive, user-facing application. CryptoAgent aims to become an Intelligent decision-making assistant for cryptocurrency portfolio optimization. In a real crypto trading team, a large amount of data is collected, including price fluctuations, trade volumes, historical performance, and news sentiment. The data scientist will then utilize advanced tools and algorithms to identify trends and forecast market

movements. In the context of our FinTech application, we conceptualize our backend models not as monolithic blocks of code, but as a team of specialist agents, simulating the structure of a real trading company. (TradingAgents-AI 2024) We define four distinct agent roles within our framework, each with a specific goal, a set of constraints, a predefined context, a unique set of skills, and access to the necessary data and models to perform its function. For instance, a sentiment analyst is equipped with tools such as search engines, including the CoinDesk API, X/twitter API, and our sentiment score system. We aim to systematically process different types of information in parallel and then synthesize them into a single, coherent investment strategy. (TradingAgents-AI 2024)

4.1 Target Users

CryptoAgent’s target users are retail crypto investors and emerging crypto fund managers, as both groups are knowledgeable or unfamiliar with the crypto space, which often lacks the resources to perform large-scale data analysis.

4.2 Analyst Team

More specifically, CryptoAgent assumes the following teams:

- The quantitative analysis Agent: this agent’s role is to act as the team’s data scientist. Its primary goal is to identify statistically and non-emotional signals from historical data.
- The sentiment analysis Agent: This agent focuses on market narratives and behavioral finance; its goal is to quantify the often-irrational sentiment that heavily influences the crypto market. Further, it is equipped with our customer VADER model, which is skilled. In filtering out noise, understanding crypto-specific jargon, and outputting a “fear-greed” index, it provides a real-time measure of the market’s emotional state.
- The portfolio management agent: this agent acts as the decision-making layer of the team. Its goal is to synthesize the inputs from both the Quant Agent and the Sentiment Agent to construct the optimal investment portfolio. Moreover, it is equipped with our backtesting and strategy construction logic. For example, it may receive “buy” signals from the Quant Agent based on strong momentum, but also receive “extreme greed” from the sentiment Agent; the portfolio management agent might then reduce the portfolio size to manage the risk.
- The client advisor agent is the user-facing agent; its goals are to translate complex analyses and decisions into clear, actionable advice for users. This agent ensures that the user is not just a passive observer of data but an informed participant in the investment process.

Figure 8: CryptoAgent Team



4.3 User Interface

Our website's interaction design is intended to be straightforward. The UI was designed with a professional look, and the application is structured around four key navigational tabs: TradingPortfolioQuantitative FactorsSentiment Analysis and News.

- **Trading dashboard:** drawing inspiration from platforms like OKX and Binance, the trading dashboard provides a real-time summary of the broader market, a list of key cryptocurrency pairs (BTC/USDT, ETH/USDT, BNB/USDT, etc) each displaying its latest price, 24-hour percentage change, and green for up, red for down. At the center of the dashboard, it displays candlestick views, multiple timeframes (ranging from minutes to days), volume bars, and the ability to overlay various technical indicators, including moving averages, RSI, and Bollinger bands. Furthermore, the fear gauge is directly adjacent to the chart, allowing users to instantly see the sentiment score and predict whether the assets will be bullish or bearish. Moreover, the trade execution panel is located at the bottom left. It provides a clear buy/sell toggle, fields for entering the trade amount and price (with options for Market, Limit, and Stop-Limit orders), and a clear display of the total order value.
- **News tabs:** News page are designed to be a clean, intuitive, and powerful center for market intelligence. Consider that investors are often inundated with an overwhelming stream of news from countless sources, CryptoAgent provides a filter for the news feed by specific categories. The default view is "all", which shows the most recent impactful news, and users can easily switch to "Bitcoin", "Ethereum", and "Market" categories.
- **Portfolio tab:** The quantitative module aims to provide information on the performance and

characteristics of our different quantitative factor portfolios. First, users can view the cumulative returns of our strategy and compare them to those of the best factor and a Bitcoin buy-and-hold investment. Second, the performance snapshot module provides key metrics such as annualized return, annualized volatility, Sharpe ratio, and maximum drawdown for each of the strategies shown in the chart. Third, the client advisor agent module provides a natural language summary for this section. Users may ask any question. Further, it is designed to proactively provide a daily or weekly “Market Briefing” that integrates all available information and offers a clear, actionable recommendation.

- Sentiment analysis: is where the core output of our sentiment analysis model is presented in full detail. First, this page provides the overall market’s sentiment gauge (our 0-100 rescaled index) and the historical daily average sentiment time series chart. More importantly, consider that users might be interested in particular cryptos, we provide a granular breakdown with an individual currency sentiment analysis section. For instance, users can view specific sentiment gauges and historical trends for Bitcoin and Ethereum, which enable them to identify divergences between the broader market sentiment and the sentiment surrounding major assets, aiming to provide a deeper layer of insight for m

4.4 Customer Onboarding Strategy

In order to successfully launch CryptoAgent, first, we need to build trust, teach users how to work efficiently with CryptoAgent, and demonstrate the potential value of our product. The initial phase would launch a pilot program offering free trading mode, which allows users to simulate our model’s performance on their own portfolios without any risk. We would then engage with community beta testers and gather feedback to refine the user interface. Second, I will publish a series of blog posts, a white paper, and a video explaining concepts such as factor investing, the role of sentiment analysis, and the logic behind our agent. Following a successful pilot phase, we would initiate a gradual rollout for real-money investment, inviting only the most engaged beta testers with a strict capital limit (e.g., a maximum of \$10,000 per user). During this stage, we do a final check of the alignment between backtested expectations and real-world results before opening the platform to a wider audience. Once we are operational, we will implement customer support to answer users’ questions, as well as enhance the stability of our backend infrastructure for our market data and news processing systems. Therefore, we aim to build a reliable and trustworthy platform for our clients in the long term.

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